

Progress Test 1.C — Thermo-S1

1.C · Equations of state and state functions

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the pressure $p = 1013 \text{ hPa}$ into bar. Give the exact result.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Equation of state and state function — definitions

(5 marks)

Define an equation of state. Write the total differential dV of the state function $V(T, p)$. Give the meaning of the partial derivative $\left(\frac{\partial V}{\partial p}\right)_T$, and its geometric interpretation in a state diagram.

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. State diagrams

(4 marks)

The curve $V(p)$ measured at constant temperature is called:

- ☐ an isotherm ☐ an isobar ☐ an isochore ☐ an adiabat

For a low-density gas, $\left(\frac{\partial V}{\partial p}\right)_T$ is:

- ☐ positive ☐ negative ☐ zero ☐ infinite

On the curve $V(T)$ at constant p , $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- ☐ the slope of the curve ☐ the area under the curve
☐ the intercept at $T = 0$ ☐ the value of V

An equation of state is valid:

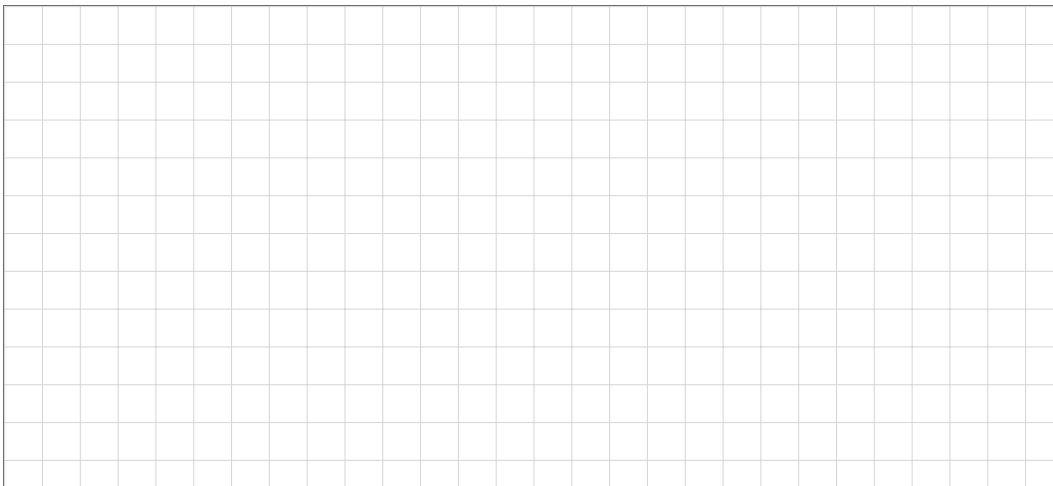
- ☐ during any transformation ☐ only at thermodynamic equilibrium
☐ only for ideal gases ☐ only at 0 °C

Exercise 4. Partial derivatives of a state function

(4 marks)

A gas obeys the equation of state $V = nRT/p$, which is given here. Compute $\left(\frac{\partial V}{\partial T}\right)_p$ and $\left(\frac{\partial V}{\partial p}\right)_T$ in the state below. Then compute the variation dV when T increases by dT and p increases by dp at the same time. Comment.

$n = 1.0 \text{ mol}$	$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$	$T = 300 \text{ K}$	$p = 1.0 \times 10^5 \text{ Pa}$
$dT = 3.0 \text{ K}$		$dp = 1.0 \times 10^3 \text{ Pa}$	

**Marking**

①	
0	1
②	
0	1
③	
0	1
④	
0	1

— End of the test —